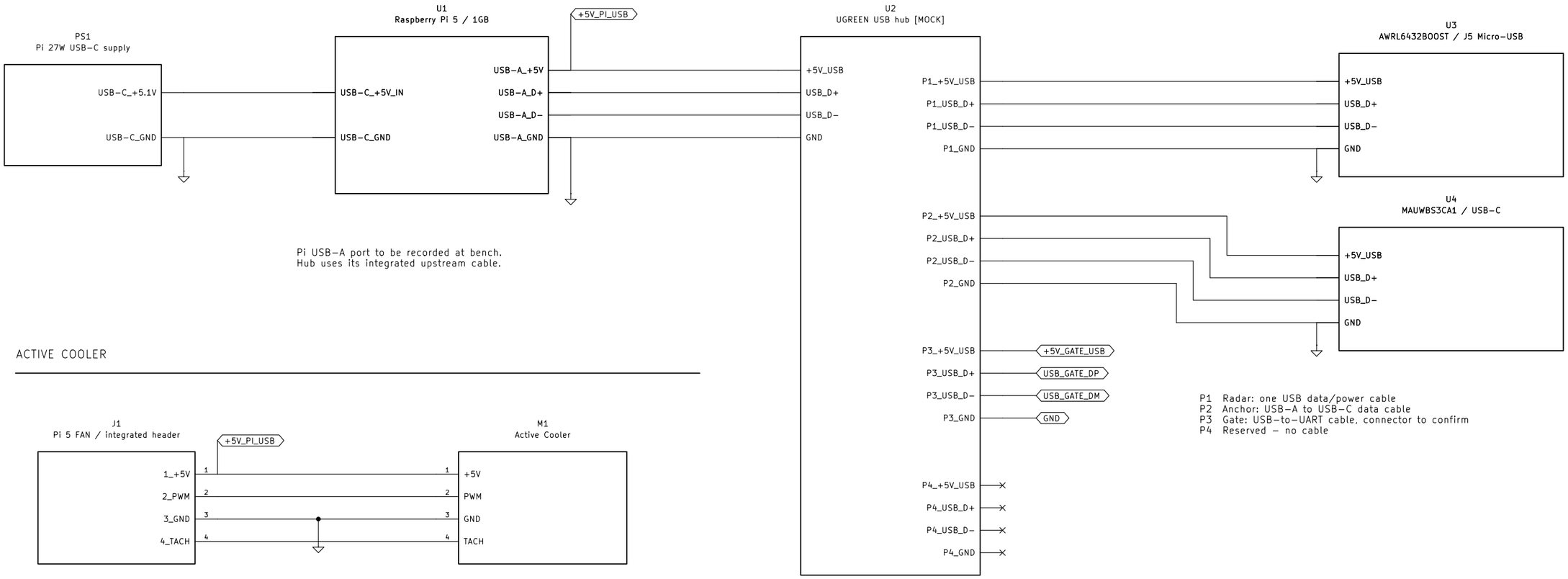
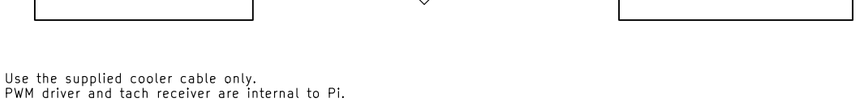
POLE UNIT / POWER AND WIRED INTERFACES



Pi USB-A port to be recorded at bench.
Hub uses its integrated upstream cable.

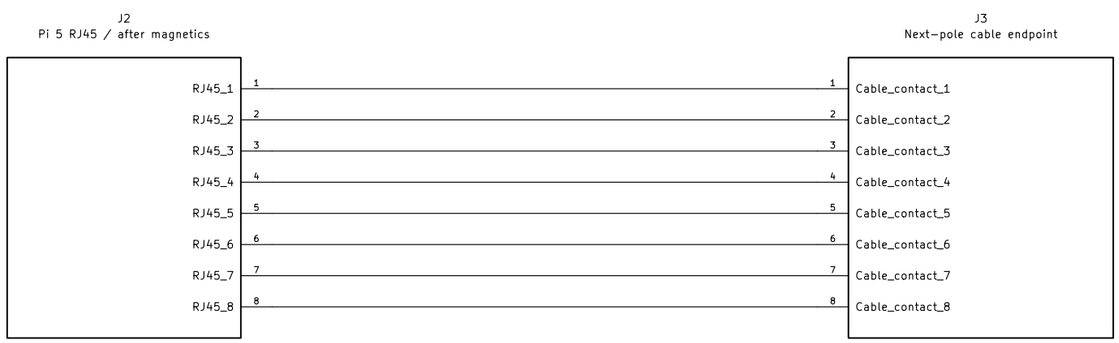
P1 Radar: one USB data/power cable
P2 Anchor: USB-A to USB-C data cable
P3 Gate: USB-to-UART cable, connector to confirm
P4 Reserved - no cable

ACTIVE COOLER



Use the supplied cooler cable only.
PWM driver and tach receiver are internal to Pi.

POLE-TO-POLE ETHERNET



CAT5e/6 straight cable. No PoE or inter-pole ground conductor.

INSTALLATION NOTES

1. Power Pi through USB-C only; prevent hub backfeed.
2. USB budget: 1.6A with 5A supply, otherwise 600mA.
   Cooler shares this allowance; measure total loading.
3. USB returns bond Pi, hub, radar, anchor and gate.
4. Hub model and self/bus power mode remain OPEN.
   [MOCK] marks the intended interface, not a verified pinout.
5. Radar high-power profiles need a measured USB budget.

Gate and wireless contract

File: Gate and Wireless.kicad_sch

DEMO ONLY | Interfaces and open items: project notes
Power, USB and wired interfaces
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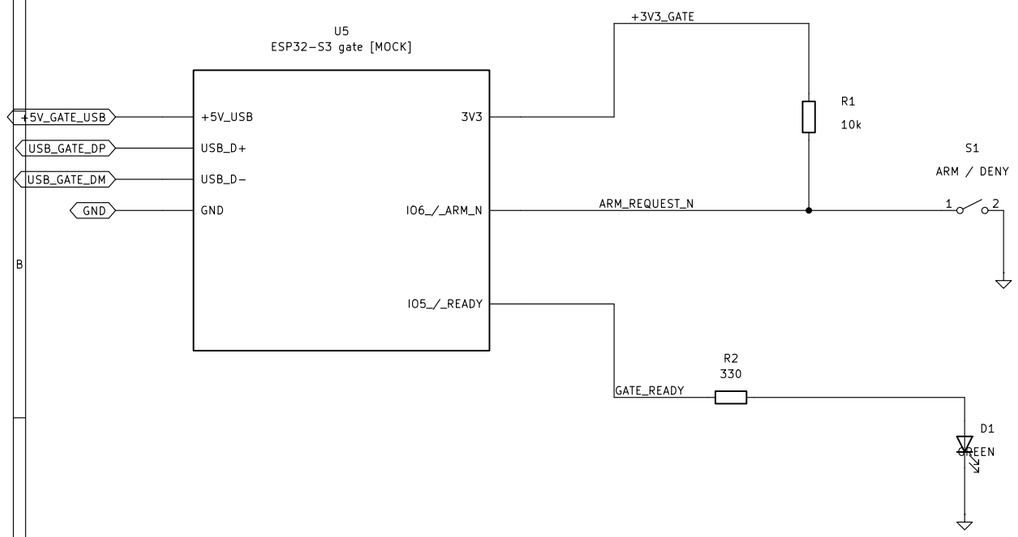
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File: Pole Unit Electronic Schematic.kicad_sch

Title: Pole Unit Electronic Schematic

Size: A3	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/2

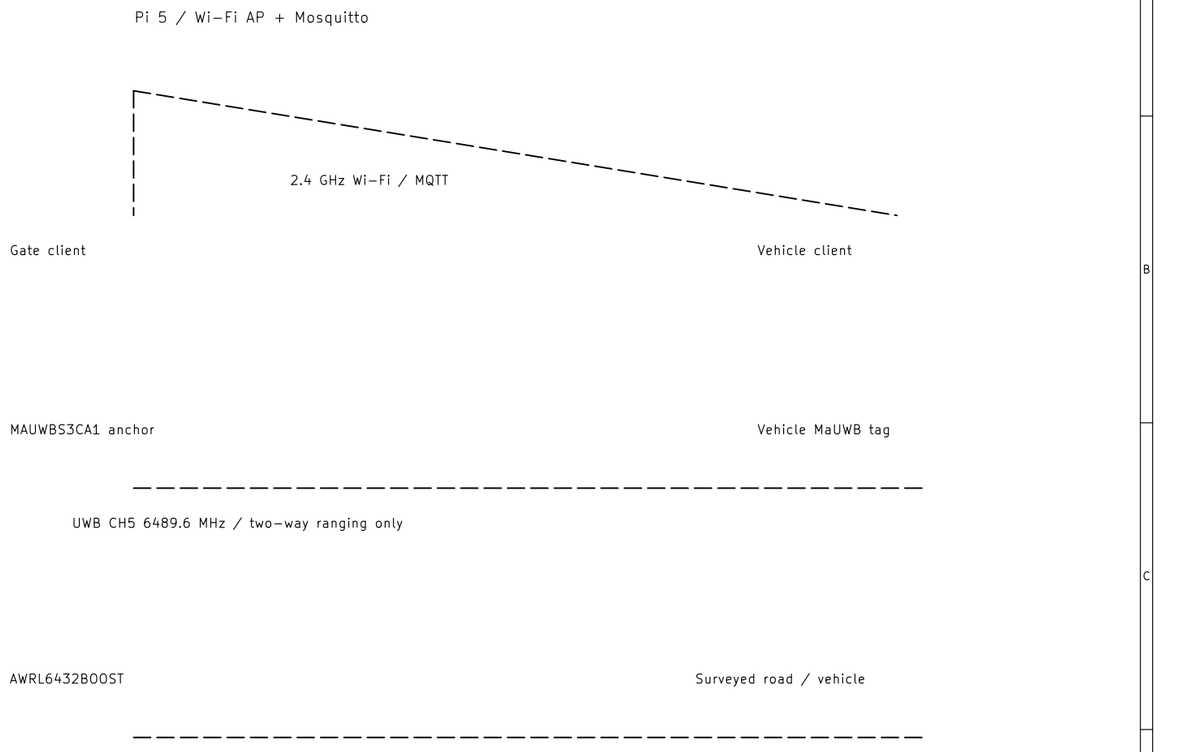
POLE UNIT / SAFETY GATE AND RADIO LINKS

3.3V GATE CIRCUIT



S1 maintained: closed = ARM request; open/broken = DENY.
Startup state: DENY. A short to GND is not detected here.
USB powers the board; no 5V connection to IO5 or IO6.

WIRELESS LINKS – LOGICAL ONLY



COMMAND INTERLOCK

PROPOSED PROTOCOL – implementation and injection test pending.
Pi sends candidate over USB; gate makes a second decision.
Gate publishes vorra/v1/vehicles/<id>/commands via Wi-Fi.
Pi owns sequence; gate preserves it and signs the full payload.
Vehicle checks gate signature, ID, session, epoch, sequence, expiry.
ACL blocks Pi-app writes to commands; key stays on gate/vehicle.
Gate shares Pi power, AP and broker: decision independence only.
DENY, malformed input or watchdog expiry: no fresh publish.

DEMO SAFETY / OPERATING NOTES

UNKNOWN, PREDICTED, stale or link loss: lamp HOLDS; horn suppression RELEASES (fail-open).
Reject expired, duplicate, retained and replayed commands. Prediction never overrides fail-open.
Physical driver override takes electrical priority and is logged. Demo hardware only; no vehicle wiring.
Radar: 57-64 GHz, 2 TX / 3 RX, 120 deg azimuth / 80 deg elevation. Multiple XDS110 ports use one cable.
No cameras, video, faces, number plates, immobilisation or V-Gun actuation. Closed-course demo only.

DEMO ONLY | Interfaces and open items: project notes
Gate, wireless links and command interlock
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V-ORRA | ITM University Gwalior

Sheet: /Gate and wireless contract/
File: Gate and Wireless.kicad_sch

Title: Pole Unit Electronic Schematic

Size: A3	Date: 2026-10-24	**Rev: A**
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